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Palaeomagnetism and geochronology of Oligocene and Miocene volcanic sections from Ethiopia: geomagnetic variability in the Afro-Arabian region over the past 30 Ma

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SUMMARY

We report palaeomagnetic and K–Ar geochronologic results of two volcanic sequences from Ethiopia. The Belessa section, dated around 29–30 Ma and spanning ∼1 km in thickness, is related to the Oligocene Afro-Arabian traps, whereas the ∼700-m-thick Debre Sina section was emplaced during the Miocene in two periods around 10–11 and 14–15 Ma. We sampled 67 flows of predominantly basaltic rocks near Belessa and 59 rhyolitic to trachybasaltic flows near Debre Sina. From a geodynamic viewpoint, the magnetostratigraphy of the Belessa sequence confirms that the Ethiopian traps were emplaced at a minimum rate of one meter per kyr, with a possible acceleration of the volume of volcanism over time. To provide insight into the evolution of the geomagnetic field in the Afro-Arabian region over the past 30 Myr, we combined our results with previous studies in the same area. Recentred directional distributions were elongated in the meridian plane, in coherence with field models for a dipole-dominated field. The dispersion S of the virtual geomagnetic poles, representative of the vigour of palaeosecular variation, was approximately 50 per cent higher during the 10–30 Ma interval than during the past 5 Myr. As the reversal frequency f was more than two times lower during the Early Oligocene than during the Plio-Pleistocene, it appears that S and f are uncorrelated in this near-equatorial region. It remains an open question whether this apparent decoupling is ascribable to a local anomaly, is only sporadic in time, or represents a general feature of the geodynamo.

Key words: Palaeomagnetic secular variation; Palaeomagnetism; Reversals: process, timescale, magnetostratigraphy.

1 INTRODUCTION

The Earth's magnetic field, generated by a dynamo process in the outer core, changes over a wide range of timescales, the most striking feature being polarity reversals at a rate of approximately four times per Myr during the past 40 Myr. An open question concerns whether the dynamo operated in different modes through geologic time. The rate of change of the field during stable periods, called palaeosecular variation (PSV), is commonly used as an indicator to distinguish among the regimes. Whereas PSV can be reconstructed using spherical harmonic models for the past millennia (e.g. Constable *et al.* 2016), PSV for older times can only be accessed with the aid of proxies extracted from the rock record (e.g. McFadden *et al.* 1988).

The most commonly used PSV proxy is the angular standard deviation S of the virtual geomagnetic poles (also known as VGP scatter) calculated from a population of individual poles derived from several sites (Cox 1970). Despite its wide use in palaeomagnetism, this proxy suffers several limitations: (1) VGP distributions are not always Fisherian-distributed (e.g. Tauxe & Kent 2004); (2) VGP distributions vary with site latitude (e.g. Cromwell *et al.* 2018) and (3) Outcrop conditions must be favourable to access a large number (typically $N > 25$) of ideally time-independent horizons. More robust estimators (i.e. better describing the tails of the distributions) can in principle be used (e.g. Suttie *et al.* 2015), but they do not circumvent the problem of latitudinal dependency. As a consequence, estimates from several latitudes are needed to fully describe the geodynamo activity at a given time. As a corollary, estimates from different time periods can only be compared if the flows were emplaced at the same palaeolatitude. Another PSV proxy, the relative variability in palaeointensity (standard deviation normalized by the average value), correlates well with the geodynamo state and is almost independent of latitude (e.g. Lhuillier & Gilder 2013). Its estimation is nevertheless challenging due to the low success rate

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of absolute or relative palaeointensity methods (e.g. Lhuillier *et al.* 2016, 2017).

Ethiopia is an ideal location to investigate the PSV evolution during the Caenozoic due to its rich volcanic activity (e.g. Kieffer *et al.* 2004). The Ethiopian traps originated at the junction of three rifts: two oceanic rifts (the Red Sea and the Gulf of Aden) and the East-African continental rift. Preceding extension, the arrival of the Afar mantle plume under the African lithosphere approximately 30 Myr ago triggered the emplacement of a *ca*. 2-km-thick pile of continental flood basalts over an area in excess of 600 000 km^2 (e.g. Hofmann *et al.* 1997). It is thought that the major part of this thick lava pile was emplaced over a 1–2 Myr interval (e.g. Hofmann *et al.* 1997; Rochette *et al.* 1998), coeval with the emplacement of the Yemeni traps (e.g. Coulié *et al.* 2003; Riisager *et al.* 2005). Fissural volcanism in Ethiopia was followed by less voluminous basaltic lavas erupted from shield volcanoes up to several tens of kilometres in basal diameter (e.g. Kieffer *et al.* 2004). Rift-related volcanism continued during the Miocene and Plio-Pleistocene and persists today.

Several palaeomagnetic studies were conducted in Ethiopia to investigate PSV. Rochette *et al.* (1998) (see also Hofmann *et al.* 1997) investigated two sections in the Oligocene flood basalts—Lima-Limo (1950 m thick) and Wegel Tena (300 m thick). The authors found an S value on the order of 15°. The overall mean pole was far-sided by *ca*. 6° with respect to the reference pole of Africa (Besse & Courtillot 2002, 2003), suggesting a quadrupolar component of the total field around 15 per cent . In a composite section of the coeval Yemeni traps, Riisager *et al.* (2005) sampled eight distinct localities over 2500 km^2 and found a similar amount of VGP scatter, but challenged the idea of a strong non-dipolar component during the Oligocene based on a global analysis. Palaeomagnetic studies on the Plio-Pleistocene Afar stratoid basalts (Kidane *et al.* 1999; Muluneh *et al.* 2013; Ahn *et al.* 2016) and in the East African rift (Kidane *et al.* 2009, 2010; Nugsse *et al.* 2018) typically yield VGP scatter on the order of 10°. A compilation of the palaeomagnetic poles from the stratoid basalts (Kidane *et al.* 2003) suggested a quadrupolar component in excess of 5 per cent .

The aim of this paper is to investigate two new sections of lava flows in Ethiopia: an Early Oligocene one near Belessa to compare against previous studies on the Afro-Arabian flood basalts and a Middle Miocene one near Debre Sina to provide comparative data on a hitherto unexplored time window in this area. Section 2 describes the sampling procedure and the geological setting while Sections 3 and 4 report the palaeomagnetic and geochronology results. Section 5 provides an analysis of the Earth's magnetic field over the past 30 Myr for the Afro-Arabian region.

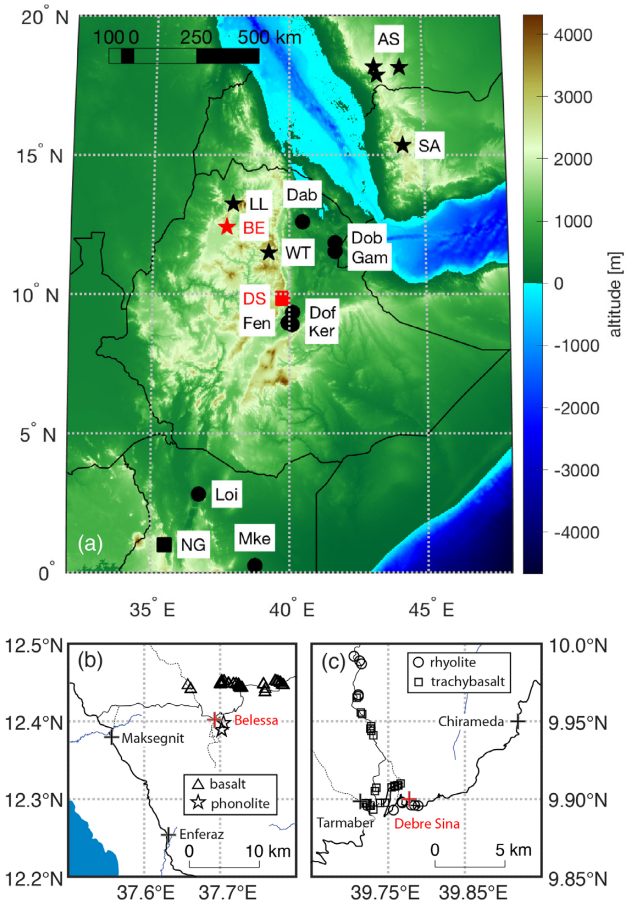


Figure 1. Panel (a): Digital elevation map (ETOPO1) of the investigated area. The coloured symbols show the positions of the present (red) and previous (black) studies. The stars represent the Early Oligocene sections (LL=Lima Limo, BE=Belessa, WT=Wegel Tena, SA=Sanaa, AS=As Sarat), the squares the Middle Miocene sections (DS=Debre Sina, NG=Ngorora), the circles the Plio-Pleistocene sections (Dab=Dabbahu, Dob=Dobi, Gam=Gamarri, Dof=Dofan, Fen=Fentele, Ker=Kereyou, Loi=Loiyangalani, Mke=Mount Kenya). Panels (b)–(c): Detailed maps of the Belessa and Debre Sina sections sampled in this study.

2 GEOLOGY AND SAMPLING

2.1 The Oligocene Belessa section

The Belessa section is located on the northwestern Ethiopian plateau near the village of Belessa (12°24′07″N, 37°41′32″E, 2450 m), approximately 30 km away from the town of Maksegnit on the Bahir Dar to Gondar motorway (Figs 1 and 2). It contains a *ca*. 1-km-thick pile of Oligocene flood basalts, mapped as TV1 to TV3 on the 1:250 000 geological map of the Yifag area (Mariam *et al.* 2012). We sampled 63 sites in the lower 800 m of the section with an average stratigraphic distance of 12 m, and six sites (due to the absence of suitable exposures) in the upper 200 m of the section with an average spacing of 50 m (Fig. 3). Our strategy was to sample all stratigraphically distinct lava flows that we could find, irrespective of the presence of palaeosols. We drilled on average eight cores per site, preferably on two or three distinct blocks near the bottom part of each flow. Of the 563 samples we collected, 542 were oriented by sun compass readings in addition to magnetic compass readings. GPS coordinates were taken at each site using a Topcon HiPerPro receiver in differential mode (base + rover) offering subdecimetre horizontal and vertical precision.

Sites B01–B10 and B12–B19 were collected in transects following dry river beds near the Belessa East road. Sites B12–B14 have a distinctive porphyritic texture that may correspond to a laccolith. Sites B20–63 were sampled in a continuous transect by hiking along a river valley and then up a ridge. Sites B64–B66 (12°23′30″N, 37°42′10″ E, 2560 m) were sampled southeast of Belessa in thick SiO_2 intermediate flows. Sites B67–B69 were sampled along a road going to the town of Wizaba. Chemical analyses of the major elements indicate that most of the sampled flows are tholeiitic basalt (Fig. 4; Table S1), with exception of Site B23 (picrobasalt) and Sites B64–66 (phonolite).

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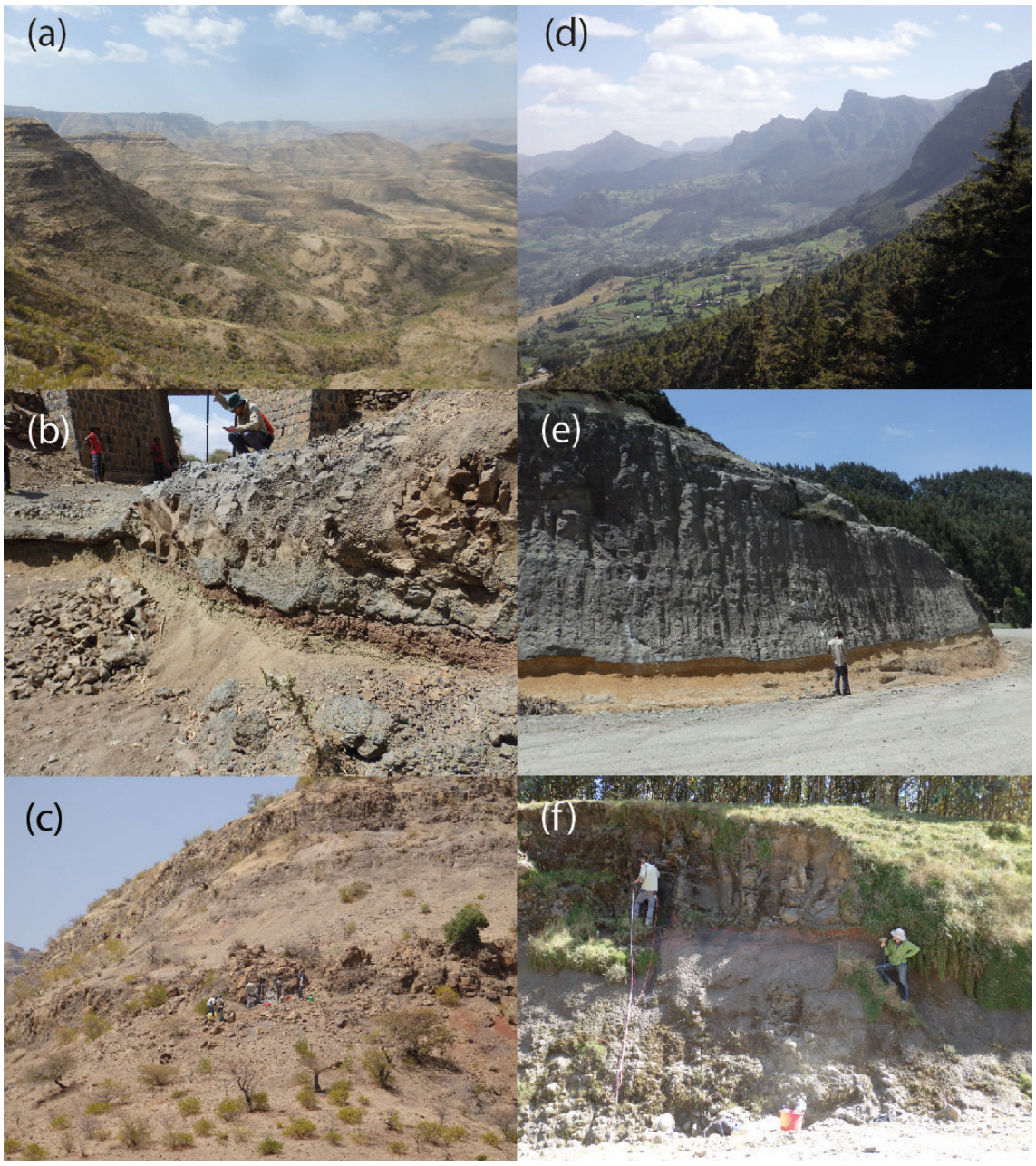


Figure 2. Field photographs of the two investigated locations. Panel (a): overview of the Early Oligocene Belessa section. Panels (b)–(c): Sites B06 and B35 showing the contact between basaltic flows. Panel (d): overview of the Miocene Debre Sina section. Panel (e): view near Site M15 showing the contact between rhyolitic ignimbrites. Panel (f): Site M42 showing the contact between trachybasaltic flows.

Field and GPS observations suggest that the flows dip *ca*. 1° to the west (~260°). Accounting for the dip is necessary to properly interpret the stratigraphic levels of the sites and avoid duplicating/overlapping flows. No tilt correction was applied to the directional data as it is negligible.

2.2 Debre Sina section

The Debre Sina section lies in the northwestern Ethiopian plateau near the city of Debre Sina (9°51′00″N, 39°45′46″E, 2700 m) on the Addis-Abeba to Dessie motorway (Figs 1 and 2). It forms part of the Tarmaber-Megezez Formation, which consists of a succession of basalts and ignimbrites/tuffs, often separated by reddish palaeosols (Meshesha *et al.* 2010). The section spans *ca*. 700 m in thickness between 2600 and 3300 m altitude (Fig. 5). We sampled 41 sites in the upper 350 m (2950–3300 m) of the section with an average stratigraphic distance of 9 m between sites, and less densely towards the base 350 m (2600–2950 m) with an average stratigraphic distance of 19 m between sites (18 sites). As for the Belessa section, we drilled on average eight cores per site, preferably in two or three distinct blocks towards the bottom part of each flow. Of the 502 samples we collected, 479 were oriented with sun compass readings

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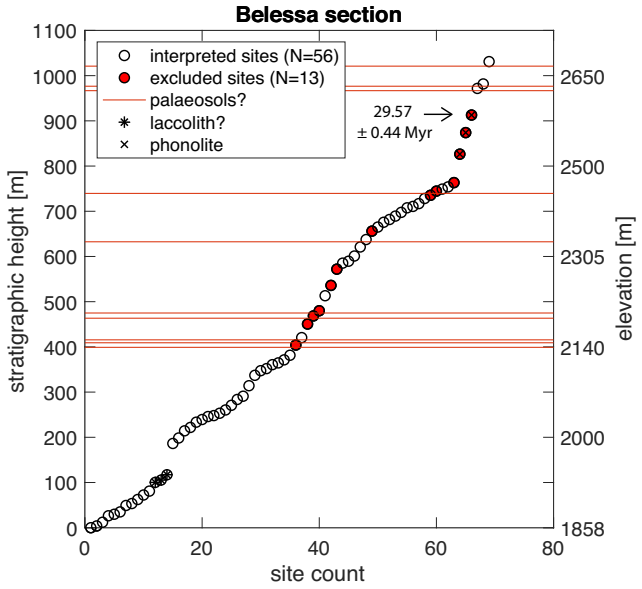


Figure 3. Stratigraphy of the Belessa section reconstructed from GPS data assuming a 1° bedding dip in the 260° direction.

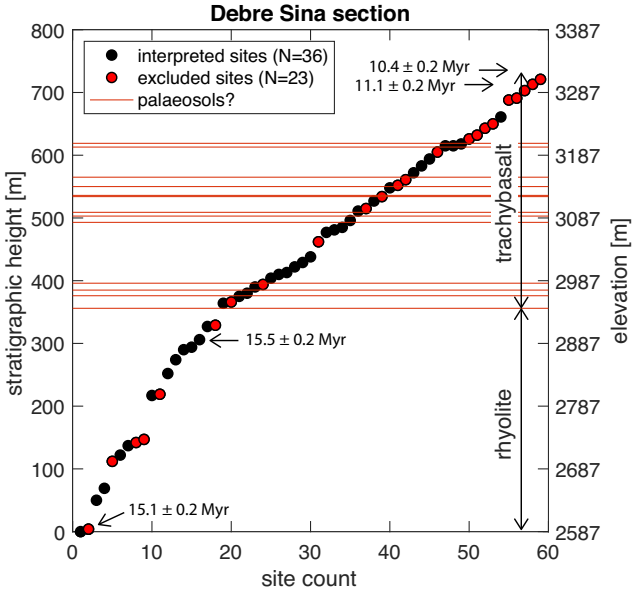


Figure 5. Stratigraphy of the Debre Sina section derived from GPS data. We assumed horizontally stratified layers.

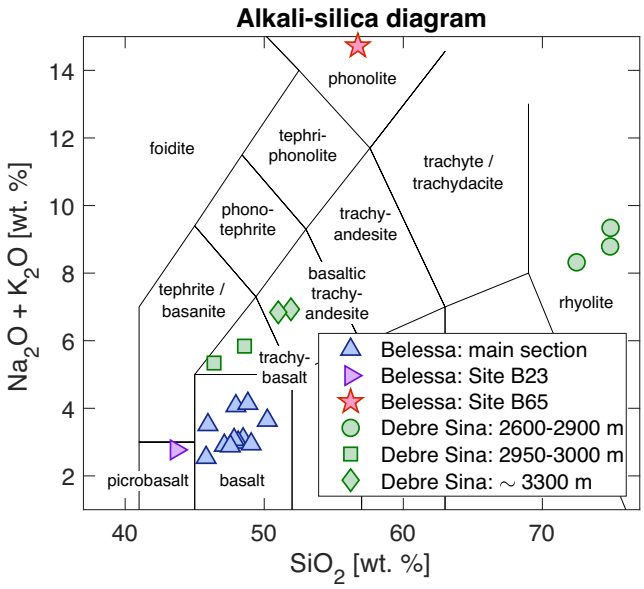


Figure 4. Chemical identification of the investigated rocks according to the total alkali-silica classification (Le Bas *et al.* 1986).

in addition to magnetic compass readings. GPS coordinates were obtained for each site with the Topcon HiPerPro receiver operating in standard mode (base only) offering submetre horizontal and vertical precision.

Sites M04–07 were collected along the road below Debre Sina, Sites M45–49 inside Debre Sina, and Sites 50–59 along the road between Debre Sina and the Tarmaber tunnel (9°50′48″N, 39°44′45″E, 3100 m). The other sites were sampled on the Mezezo road off the Tarmaber pass. Each site corresponds to an independent volcanic flow or ignimbrite/tuff layer and we assumed that the sequence was horizontally stratified. Chemical analyses of the major elements indicate that the lower part of the section (2600–2950 m) is rhyolitic, whereas the upper part (2950–3300 m) is more basic, with trachybasalts towards the bottom and basaltic trachyandesites towards the top (Fig. 4, Table S1); such a bimodal geochemical distribution is typical of rift environments (e.g. Olsen 1995).

3 PALAEOMAGNETIC DIRECTIONS

3.1 Palaeomagnetic experiments

Magnetic susceptibility χ_m was measured before demagnetization using a Bartington MS2B meter operating at a frequency of 0.465 kHz. The Koenigsberger ratio Q, defined as the natural remanent magnetization (NRM) divided by the induced magnetization ($\chi_m \cdot B/\mu_0$), was computed for a present-day magnetic field $B = 37$ μT at Belessa and $B = 36$ μT at Debre Sina (Thébault *et al.* 2015), where μ_0 is the magnetic constant.

Thermal demagnetization was carried out using an ASC TD-48 furnace together with an AGICO JR6 spinner magnetometer. Alternating field (AF) demagnetization up to 90 mT was done using the automated SushiBar system based on a three-axis 2G Enterprises superconducting magnetometer and a custom-made coil (Wack & Gilder 2012). The specimen-level characteristic remanent magnetization (ChRM) directions were determined using principal component analysis (PCA, Kirschvink 1980) with the help of the paleomagnetism.org environment (Koymans *et al.* 2016). For the Belessa section, we AF (resp. thermally) demagnetized 550 (resp. 250) specimens. For the Debre Sina section, we AF (resp. thermally) demagnetized 340 (resp. 250) specimens. The median destructive field (MDF)—the value for which half of the initial natural remanent magnetization (NRM) is randomized—was computed for each sample from the decay curve reconstructed by modular subtraction.

From the site-mean directions, we calculated the Fisher (1953) statistics (precision parameter k, 95 per cent confidence radius α_{95},

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angular standard deviation s) and the elongation parameter e (Tauxe 2010). To assess the serial correlation between adjacent lava flows, we used the method proposed by Chenet *et al.* (2008). The angular distance δ between the site-mean directions of each pair of adjacent lava flows was compared to a threshold value δ_c given by the quadratic sum of the corresponding α_{95} values. When δ does not exceed δ_0, the mean directions of the adjacent flows are hypothetically correlated and their individual directions were combined into a single directional group. For the two investigated sections, we systematically tested the potential correlation of each pair of consecutive lava flows, irrespective of the presence of palaeosols.

Magneto-mineralogies of the samples were investigated by measuring continuous thermomagnetic curves for one or two samples per site on a variable field translation balance (VFTB), while heating in air from room temperature to 600 °C in a field of approximately 200 mT. The dominant Curie point T_C was determined as the temperature corresponding to the absolute minimum of the first-time derivative (e.g. Fabian *et al.* 2013).

3.2 Belessa section

3.2.1 Site selection

Fig. 6 presents the site-mean magnetic anomalies between declination-corrected magnetic azimuths and sun azimuths (panel a), the site-mean NRM values (panel b) and the site-mean Koenigsberger ratios (panel c). Eight sites (B36, B38, B40, B42, B43, B49, B59 and B60) had both strong magnetic anomalies (from 10° to 50°) and high Koenigsberger ratios (from 20 to 200). We suspect that most of these sites were lightning-struck and consequently excluded them from subsequent analyses. Sites B39, B63 and B64–66 (the latter corresponding to the phonolitic flows) yielded uninterpretable demagnetization behaviour and were also excluded, leaving 56 sites to base our investigation. The average magnetic anomaly of the 56 sites is −0.2°. NRM values range between 0.4 and 24 A m^{-1} with a median at 1.7 A m^{-1}, in agreement with previous studies of the Ethiopian traps (e.g. Rochette *et al.* 1998; Kidane *et al.* 2002). The Koenigsberger ratio has a median value of 4.7.

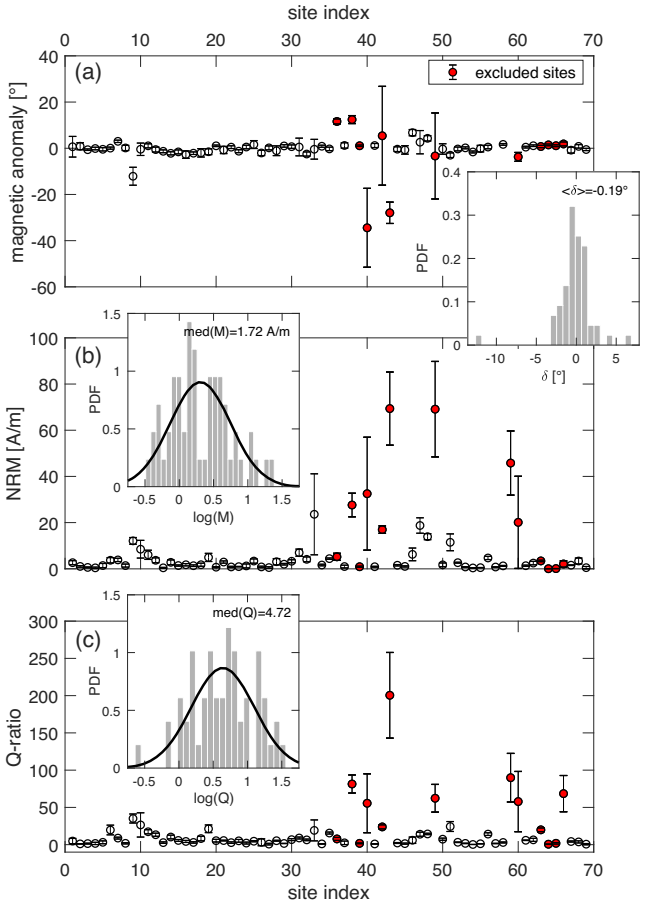


Figure 6. Site selection for the Belessa section. Panel (a): site-mean magnetic anomaly between declination-corrected magnetic azimuths and sun azimuths. Panel (b): site-mean NRM values. Panel (c): site-mean Koenigsberger ratios. Only the accepted sites are presented in the histograms.

3.2.2 Demagnetization behaviour

Fig. 7 shows typical behaviour of thermal and AF demagnetized sister specimens for a site of normal polarity (B06), for a site of reverse polarity (B28) and for a site with a transitional direction (B68). Sample B28-229 is representative of the majority of the sites—namely a secondary component removed by 20–30 mT or 200–300 °C, followed by a unidirectional decay of the vector endpoints to the origin upon further demagnetization. ChRM directions were determined with origin-anchored PCA fits. With the exception of Site B46, AF demagnetization results were, for the sake of homogeneity, favoured to compute the site-mean directions. Site-mean directions yielded α_{95} values ranging from 1° to 9° with an average of 3.8° (Table 1).

High-field thermomagnetic curves (Fig. S1) reveal a wide range of Curie temperatures from 100 to 600 °C, and sometimes the coexistence of two ferrimagnetic phases (e.g. specimen B06-041T in Fig. 7), consistent with the presence of titanomagnetite. The dominant Curie points are above 450 °C in 50 per cent of the cases and lower than 200 °C in 20 per cent of the cases. The MDF is higher than 16 mT in 50 per cent of the cases, and lower than 7 mT in 20 per cent of the cases. Note that this quantity must be interpreted with caution in case of multiple demagnatization components. The values below 10 mT for this section do not reflect low coercivities but rather correspond to the presence of strong secondary components probably caused by lightning.

3.2.3 Directional groups and PSV statistics

We identified 21 potentially correlated flows, which were combined into 10 directional groups (reported in italics in Table 1). Among this set of 45 independent directions, 23 are of normal polarity. The Vandamme (1994) cut-off procedure identified three transitional directions (Sites B01, B51 and B68), highlighted in red in Fig. 8. The between-site dispersion S_B of stable VGPs, corrected for within-site dispersion (e.g. Johnson *et al.* 2008), yields $10.3^{\circ}|^{13.3^{\circ}}_{8.4^{\circ}}$ with $N = 20$ for the normal polarities, $11.4^{\circ}|^{14.5^{\circ}}_{9.4^{\circ}}$ with $N = 22$ for the reverse polarities, $12.1^{\circ}|^{14.3^{\circ}}_{10.5^{\circ}}$ with $N = 42$ for the combined polarities. The angle between the normal directions and reverse directions 11.1° exceeds the critical angle 8.2° for which the two populations would be different at the 95 per cent significance level, so the reversal test is negative. This result remains unchanged when performing the reversal test after discarding the sites with median destructive fields lower than 10 mT. We note that Sites B12-14, potentially identified as a laccolith from field observations, yields directions that cannot

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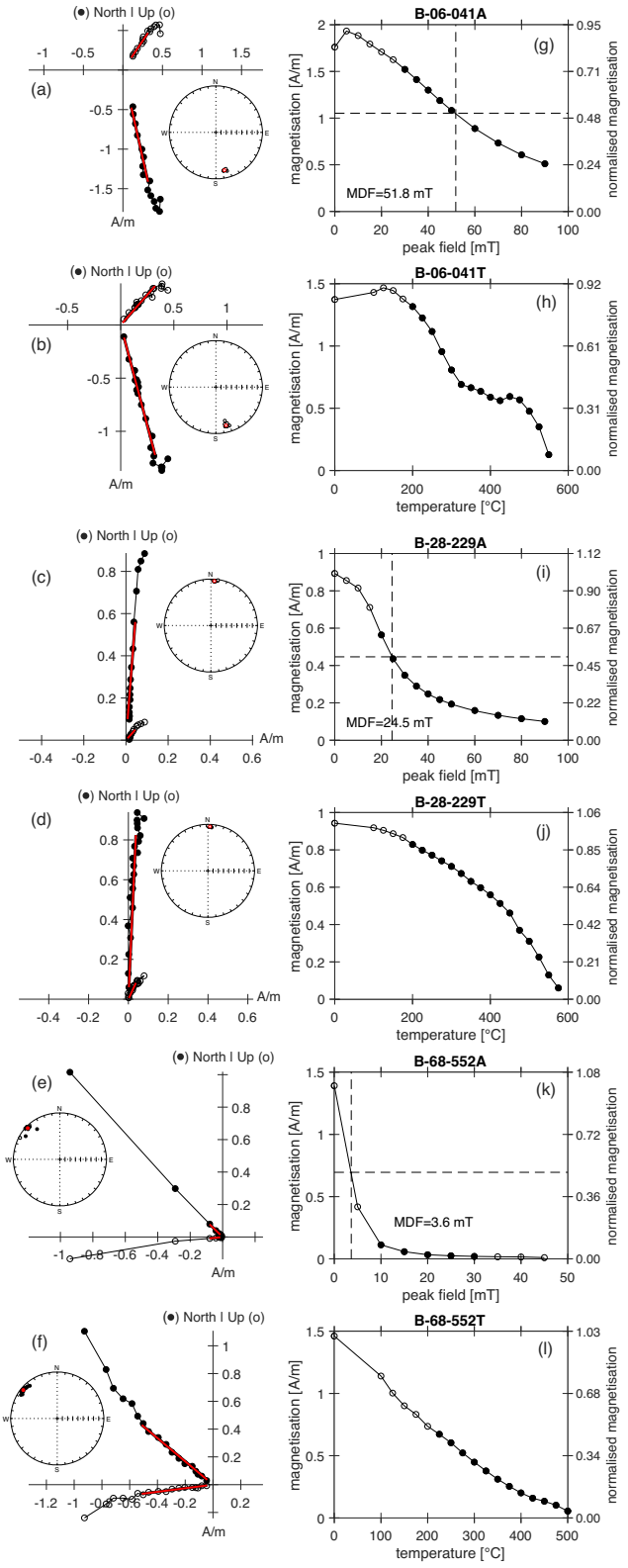


Figure 7. Typical behaviour of thermal and AF demagnetized sister specimens for the Belessa section. Panels (a)–(f): orthogonal plots with corresponding stereonets. Panels (g)–(l): decay plots. The median destructive field (MDF) corresponds to the AF field for which half of the initial NRM is randomized. The PCA directions are shown in red.

be distinguished from those of the rest of the section, which suggests that they may rather be ordinary lava flows though with a distinctive porphyritic texture.

3.3 Debre Sina section

3.3.1 Site selection

Fig. 9 presents the site-mean magnetic anomalies, that is the angle between declination-corrected magnetic azimuths and sun azimuths (panel a), the site-mean NRM values (panel b) and the site-mean Koenigsberger ratios (panel c). Sites M27, M29–33, M36, M38 and M40 were excluded because they had magnetic anomalies greater than 10°, NRMs greater than 30 A m^{-1} and Koenigsberger ratios greater than 30. We suspect that most of these sites were lightning-struck. Sites M12–13, M17, M34, M37, M45 and M50 did not show a strong magnetic anomaly but were excluded because they had NRMs greater than 20 A/m and Koenigsberger ratios greater than 10. Site M01 was excluded because it had a Koenigsberger ratio greater than 10 and had inconsistent directions between the three sampled blocks. Sites M5, M10, M43, M47, M49 were excluded because of erratic and/or ambiguous demagnetization behaviour. We disregarded the 23 anomalous sites and based our investigation on the 36 remaining sites.

Once the aberrant sites were removed, the mean magnetic anomaly is 0.2°. The NRM values range from 2 to 45 A m^{-1} with a median of 3.3 A m^{-1}, whilst the median Koenigsberger ratio is 2.1. A marked difference in magnetization characteristics is observed between the rhyolitic ignimbrites (median NRM at 26 m A m^{-1}, median Q-ratio at 0.8) and the trachybasalts (median NRM at 4.8 A m^{-1}, median Q-ratio at 2.4).

3.3.2 Demagnetization behaviour

Fig. 10 shows typical thermal and AF demagnetization behaviour of sister specimens for site (M07) at the bottom of the section, site (M57) at the top of the section and for a site identified as having a transitional direction (M48). For the ignimbrites, thermal demagnetization was usually more efficient than AF demagnetization to isolate linear magnetization components. It was the opposite for the trachybasalts. That notwithstanding, the two methods yielded in most cases unidirectional decay of the vector endpoints to the origin, so that the ChRM directions were determined with origin-anchored PCA fits. Thermal and AF demagnetization results were combined to compute the site-mean directions (Table 2). Site-mean directions yielded α_{95} values ranging from 1.6° to 10.6° with an average of 4.4°. Only one site (M39) had an α_{95} value exceeding 10°.

Like for the Belessa section, the magnetic carrier is mostly titanomagnetite, often with two ferrimagnetic phases. High-field thermomagnetic curves reveal that the dominant Curie temperature is higher than 400 °C in 40 per cent of the cases, with 40 per cent of the sites with $T_C \leq 200$ °C (Fig. S2). A few samples from ignimbrites at the bottom of the section did not fully demagnetize by 600 °C and were partially resistant to AF demagnetization, which suggests that they contain a small fraction of hematite. The MDF is higher than 21 mT in 40 per cent of the cases and lower than 11.5

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Table 1. Palaeomagnetic results for the Belessa section: site (or group) name; longitude (slon), latitude (slat) and altitude (alt) of the site; stratigraphic height (h_{strat}); number n/N of interpreted/measured specimens by thermal (TH) or alternating field (AF) demagnetization; declination (D_{geo}) and inclination (I_{geo}) in geographic coordinates with the precision parameter (k) and the 95 per cent confidence radius (α_{95}); longitude (plon) and latitude (plat) of the virtual geomagnetic poles; median destructive field (MDF) and maximum unblocking temperature T_u. Flows marked with an asterisk (∗) were merged into directional groups highlighted in italics. Flows marked with a cross (†) record excursional directions.

Site/group	Slon [°E]	Slat [°N]	Alt [m]	h_{strat} [m]	n/N(TH)	n/N(AF)	D_{geo} [°]	I_{geo} [°]	k	α_{95} [°]	Plon [°N]	Plat [°E]	MDF [mT]	T_u [°C]
B01†	37.7836	12.4463	1859	0	0/4	5/8	21.9	−19.1	92	8.0	172.4	58.7	13	300
B02	37.7825	12.4465	1860	4	0/2	8/8	176.3	−11.5	139	4.7	66.9	−82.4	25	525
B03	37.7811	12.4473	1866	12	0/8	6/8	157.4	−27.9	304	3.8	136.7	−67.8	14	450
B04∗	37.7797	12.4494	1879	26	0/4	5/8	171.4	−33.0	615	3.1	162.7	−80.0	16	550
B05∗	37.7792	12.4495	1881	30	0/4	5/8	168.4	−28.7	76	8.8	143.6	−78.2	15	525
B04–05				28	0/8	10/16	169.9	−30.9	132	4.2	−79.3	144.1		
B06	37.7786	12.4499	1885	35	0/1	8/8	168.7	−15.7	157	4.4	107.1	−78.0	36	550
B07	37.7765	12.4505	1896	49	0/3	8/8	186.1	3.1	429	2.7	13.8	−74.7	5	500
B08	37.7760	12.4507	1899	53	0/1	8/8	186.4	−2.0	253	3.5	8.1	−76.9	28	575
B09∗	37.7748	12.4507	1906	62	0/2	7/8	166.6	−7.8	197	4.3	96.1	−74.3	5	350
B10∗	37.7732	12.4517	1913	72	0/2	8/8	166.0	−8.4	160	4.4	98.3	−73.9	11	425
B09–10				67	0/4	15/16	166.3	−8.1	187	2.8	−74.1	218.1		
B11	37.7721	12.4520	1919	81	0/2	8/8	155.1	−6.2	143	4.6	109.1	−63.7	10	300
B12∗	37.7588	12.4372	1909	100	0/1	8/10	182.6	−5.0	269	3.4	22.8	−79.7	48	600
B13∗	37.7589	12.4380	1915	105	0/4	6/8	181.7	−1.3	105	6.6	29.7	−78.1	19	275
B12–13				103	0/5	14/18	182.2	−3.4	160	3.2	−79.1	208.1		
B14	37.7593	12.4416	1928	117	0/1	8/8	184.4	−8.1	348	3.0	9.8	−80.6	39	600
B15∗	37.7572	12.4473	1996	186	0/2	8/8	178.9	−20.6	1026	1.7	69.7	−87.9	28	550
B16∗	37.7571	12.4475	2008	198	0/1	8/8	177.5	−21.5	1140	1.6	99.7	−87.2	32	550
B15–16				192	0/3	16/16	178.2	−21.1	1044	1.1	−87.7	1513.9		
B17	37.7571	12.4478	2024	214	0/3	8/8	167.9	−3.6	307	3.2	87.0	−74.0	15	325
B18	37.7571	12.4479	2031	222	0/3	9/9	173.7	−12.7	175	3.9	84.0	−81.4	11	525
B19	37.7567	12.4479	2042	233	0/1	8/8	163.3	−26.7	171	4.3	135.6	−73.6	68	600
B20∗	37.7290	12.4431	1995	239	0/1	8/8	163.7	−12.4	361	2.9	108.2	−72.7	14	525
B21∗	37.7285	12.4430	2000	246	0/1	9/9	162.0	−9.5	294	3.0	106.1	−70.7	41	575
B20–21				243	0/2	17/17	162.8	−10.9	299	2.1	−71.6	482.2		
B22∗	37.7279	12.4430	2001	248	0/3	9/9	163.2	−4.4	297	3.0	97.5	−70.4	22	500
B23∗	37.7269	12.4427	2005	253	0/4	6/8	166.0	0.3	112	6.4	86.6	−71.2	30	550
B22–23				251	0/7	15/17	164.3	−2.5	159	3.0	−70.8	183.0		
B24	37.7258	12.4435	2010	260	0/3	8/8	158.5	−0.6	208	3.8	99.6	−65.4	21	550
B25∗	37.7253	12.4437	2019	271	0/6	5/8	174.5	7.5	184	5.7	56.5	−72.9	10	400
B26∗	37.7249	12.4447	2032	284	0/2	7/8	174.7	7.9	437	2.9	55.8	−72.8	35	500
B25–26				277	0/8	12/16	174.6	7.7	309	2.5	−72.8	391.3		
B27	37.7249	12.4450	2039	291	0/3	8/8	184.9	6.3	161	4.4	20.0	−73.6	23	525
B28	37.7251	12.4461	2063	314	0/1	7/8	2.1	−2.8	315	3.4	209.2	76.0	25	575
B29	37.7245	12.4467	2085	337	0/1	8/8	349.5	2.2	852	1.9	261.0	74.6	24	575
B30	37.7243	12.4468	2095	347	0/8	8/8	12.8	8.3	774	2.0	159.7	74.9	4	500
B31	37.7242	12.4469	2100	352	0/3	9/9	4.6	11.6	2023	1.1	182.8	82.0	28	550
B32	37.7241	12.4474	2108	361	0/2	7/8	9.9	14.6	921	2.0	153.9	79.0	21	550
B33∗	37.7212	12.4480	2107	365	0/5	4/8	0.2	28.0	164	7.2	43.1	87.5	16	525
B34∗	37.7209	12.4478	2114	371	0/1	8/9	7.0	24.9	80	6.2	120.6	82.9	12	525
B35∗	37.7205	12.4476	2123	381	0/4	5/9	1.4	24.0	569	3.2	123.7	88.6	5	425
B33–35				375	0/10	17/26	3.8	25.4	115	3.3	86.1	167.9		
B37	37.7172	12.4480	2156	421	0/1	9/10	3.2	−12.2	828	1.8	207.9	71.1	16	600
B41	37.7054	12.4515	2227	513	0/8	8/8	1.0	22.6	98	5.6	158.2	88.8	8	400
B44	37.7019	12.4515	2293	585	0/9	8/9	1.2	−3.3	123	5.0	212.9	75.8	5	500
B45	37.7017	12.4513	2297	590	0/8	8/8	0.2	12.1	71	6.6	216.4	83.7	9	450
B46	37.7015	12.4511	2308	601	7/8	7/8	168.9	11.7	171	4.6	69.4	−68.5	6	400
B47	37.7012	12.4507	2327	621	0/1	6/8	161.0	−3.0	198	4.8	99.0	−68.2	11	500
B48	37.7007	12.4499	2342	637	0/1	8/8	173.7	−17.3	625	2.2	98.2	−82.9	7	525
B50	37.7004	12.4493	2369	665	0/1	7/8	11.4	18.9	402	3.0	140.3	78.5	28	575
B51†	37.7005	12.4490	2380	676	0/3	8/8	27.9	−14.8	397	2.8	162.1	55.8	6	400
B52	37.7005	12.4487	2386	682	0/1	8/8	7.9	−3.8	654	2.2	188.5	73.6	19	575
B53	37.7006	12.4484	2393	689	0/1	6/8	6.0	4.7	1453	1.8	186.7	78.3	16	500
B54	37.7004	12.4484	2401	698	0/4	8/8	352.2	−13.2	249	3.5	240.3	69.3	13	525
B55∗	37.7006	12.4480	2412	708	0/1	8/8	355.6	2.1	208	3.8	239.0	77.8	7	500
B56∗	37.7006	12.4480	2415	711	0/1	8/8	356.5	0.5	437	2.7	234.0	77.3	28	600
B55–56				709	0/2	16/16	356.0	1.3	291	2.2	77.6	453.0		
B57∗	37.7005	12.4480	2421	717	0/1	8/8	0.3	4.8	494	2.5	215.8	79.9	20	525
B58∗	37.7008	12.4478	2433	728	0/1	7/9	2.3	5.9	1223	1.7	203.8	80.2	21	550
B57–58				723	0/2	15/17	1.3	5.3	634	1.5	80.1	1437.3		
B61	37.7001	12.4478	2452	749	0/1	8/8	357.2	−19.2	788	2.0	224.9	67.5	38	575
B62	37.6999	12.4479	2457	754	0/6	8/8	10.8	7.8	531	2.4	165.2	76.3	5	525
B69	37.6614	12.4410	2600	972	0/6	6/7	340.2	26.7	90	7.1	315.1	70.5	19	525

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Table 1. Continued

Site/group	Slon [°E]	Slat [°N]	Alt [m]	h_{strat} [m]	n/N(TH)	n/N(AF)	D_{geo} [°]	I_{geo} [°]	k	α_{95} [°]	Plon [°N]	Plat [°E]	MDF [mT]	T_u [°C]
B68†	37.6611	12.4416	2610	982	0/8	6/8	307.6	8.6	125	6.0	303.5	37.7	4	475
B67	37.6574	12.4463	2654	1031	0/1	8/8	8.0	3.4	843	1.9	180.7	76.7	40	575

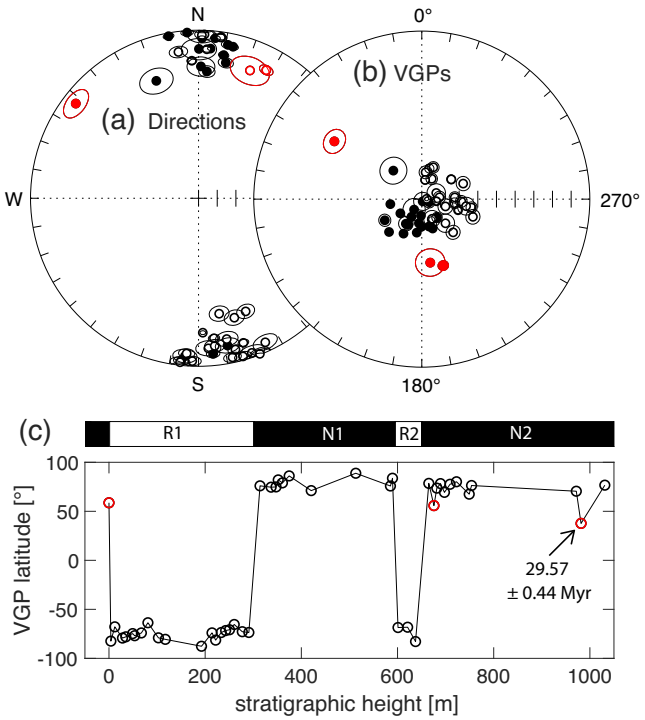


Figure 8. Site-mean directions for the Belessa section. Panel (a): dispersion of the directions. Panel (b): dispersion of the VGPs. Panel (c): magnetostratigraphy. The red symbols stand for the transitional sites.

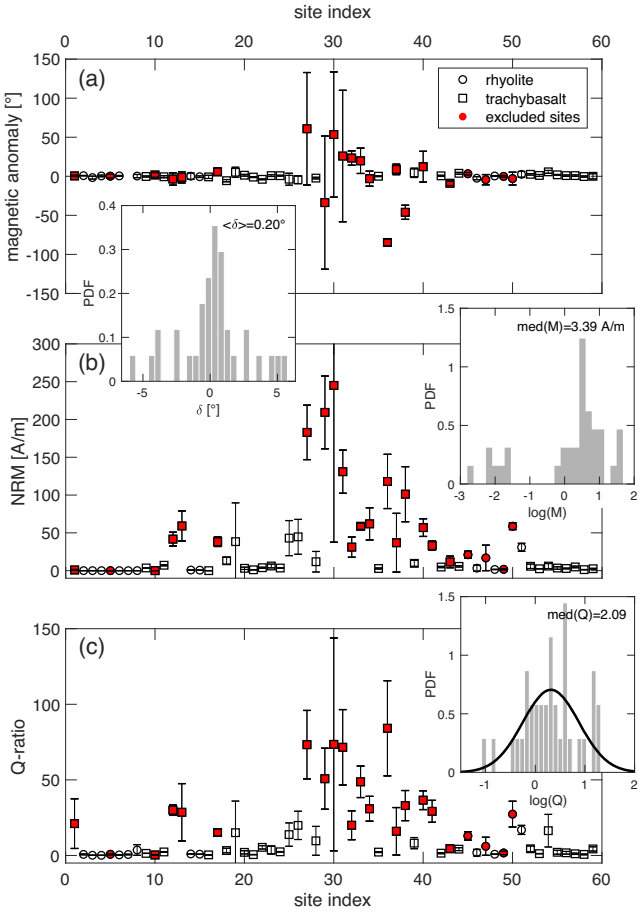


Figure 9. Site selection for the Debre Sina section. Panel (a): site-mean magnetic anomaly (the angle between declination-corrected magnetic azimuths and sun azimuths). Panel (b): site-mean NRM values. Panel (c): site-mean Koenigsberger ratios. The histograms plot only the accepted sites.

mT in 40 per cent of the cases. The values below 10 mT again correspond to the presence of strong secondary components probably induced by lightning strikes.

3.3.3 Directional groups and PSV statistics

Four potentially correlated flows (M14+M15, M28+M35) were combined into two directional groups (Table 2). Out of 34 independent directions, only three non-contiguous site-mean directions of reverse polarity were found. The Vandamme (1994) cut-off procedure identified four transitional directions (Sites M02, M18, M46 and M48), highlighted in red in Fig. 11. The between-site dispersion S_B of stable VGPs, corrected for within-site dispersion (e.g. Johnson *et al.* 2008), yields $14.2^\circ|^{17.5^\circ}_{12.0^\circ}$ with $N = 28$ for the normal polarities, $15.5^\circ|^{18.9^\circ}_{13.1^\circ}$ with $N = 30$ for the combined polarities. The reversal test is indeterminate (Ro) according to the classification of McFadden & McElhinny (1990). The lower part of the section (ignimbrite) has only eight sites with stable directions, which is insufficient to calculate a robust S_B value. The upper part of the section (trachybasalts) yields $S_B = 13.0^\circ|^{16.6^\circ}_{10.7^\circ}$, which is, at a 95 per cent confidence level, not statistically different from the estimate for the whole section.

4 GEOCHRONOLOGY

4.1 Method

At each section we collected eight, half kilogram samples at various stratigraphic levels. Determination of the loss on ignition and chemical analyses of the major elements (Table S1) revealed that only seven samples were suitable for K-Ar dating (Table 3). Radiometric analyses were carried out using the unspiked K-Ar technique (Charbit *et al.* 1998; Guillou *et al.* 2018). Argon extracted from the sample was measured in sequence with purified aliquots of atmospheric argon at comparable working gas pressure in the mass spectrometer to suppress mass discrimination effects between the atmospheric reference and the unknown. One or multiple manometrically calibrated doses of atmospheric argon were used to convert beam intensities into atomic abundances and to monitor the atmospheric correction.

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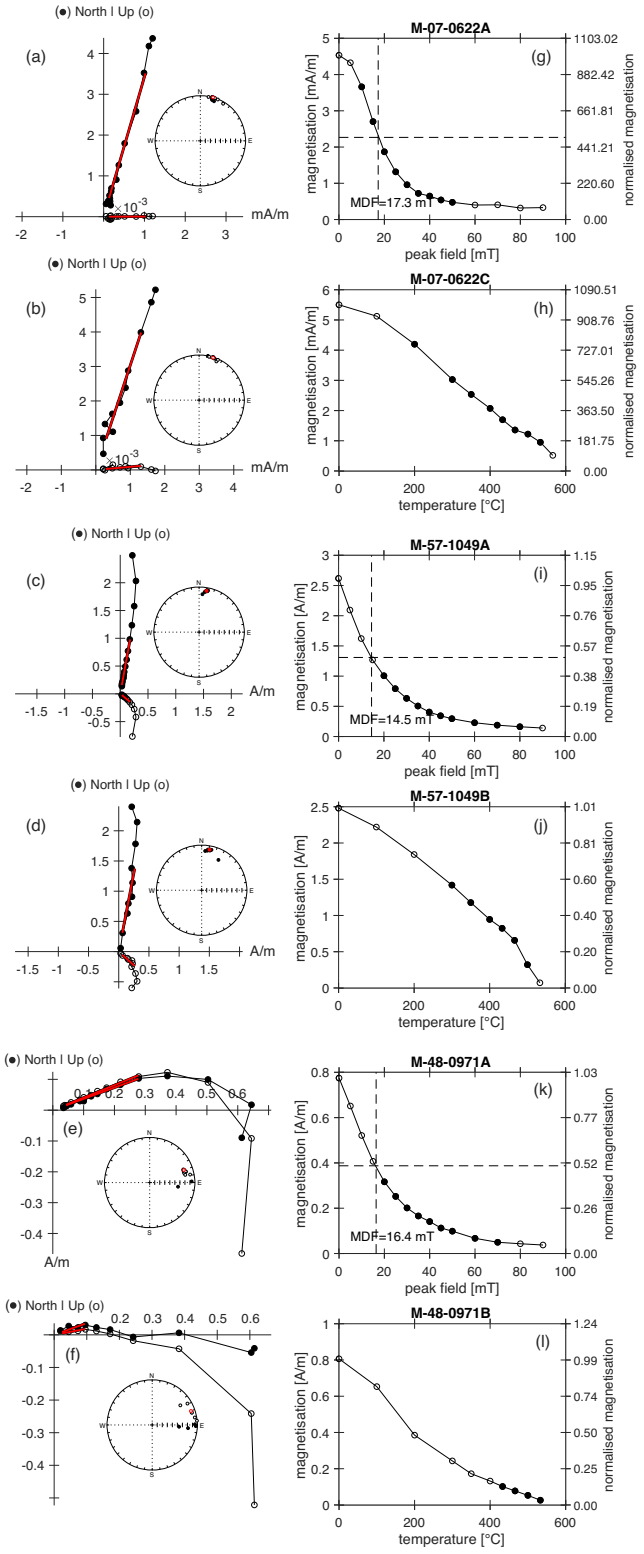


Figure 10. Typical thermal and AF demagnetization behaviour of sister specimens for the Debre Sina section. Panels (a)-(f): orthogonal plots with corresponding stereonets. Panels (g)-(l): decay plots. The median destructive field (MDF) corresponds to the AF field for which half of the initial NRM is randomized. The PCA directions are shown in red.

4.2 Results

At the Belessa section, samples B02 and B06 from the bottom of the section yielded K-Ar ages younger than sample B68 at the top of the section. From prior age constraints of the Afro-Arabian traps (e.g. Coulié *et al.* 2003), we suspect that the ages of B02 and B06 were likely underestimated. This hypothesis is also corroborated by the fact that the bottom of the section was visually and microscopically more altered than the top part, making a loss of ^{40}Ar$*$ or a gain of K in these samples likelier. As a result, we only accepted the age of 29.57 ± 0.44 Ma from site B68 and used it to correlate with the geomagnetic polarity time scale (GPTS).

For the Debre Sina section, the ages for samples M05 (15.08 ± 0.23 Ma, at 2591 m) and M02 (15.45 ± 0.23 Ma, at 2893 m) agree within uncertainty. Consistent with the stratigraphy, M33 (11.11 ± 0.17 Ma, at 3300 m) and M34 (10.40 ± 0.16 Ma, at 3308 m) near the top of the section are younger than the underlying ignimbrites. Based on these data, we suspect that a 3–4 Myr hiatus exists between the rhyolitic ignimbrites in the lower part of the section (2600–2950 m) and the trachybasalts in the upper part of the section (2950–3300 m).

4.3 Comparison with the geomagnetic polarity timescale

The magnetostratigraphy of the Belessa section (Fig. 8) yields four stable polarity periods excluding the lowermost site with normal polarity. Taking into account that the bottom of the section cannot be older than chron C12n (Coulié *et al.* 2003; Riisager *et al.* 2005), we correlated site B68, dated at 29.57 ± 0.44 Ma, with normal polarity chron C11n calibrated between 29.97 and 29.18 Ma (Ogg 2012), and in particular with subchron C11n.1n (Fig. 12). Starting from the top of the section, the last 400 m of normal polarity (N2) can be associated with subchron C11n.1n, the preceding 55 m of reverse polarity (R2) with subchron C11n.1r, and the first 300 m of reverse polarity (R1) with chron C11r. B01, the sole site with normal polarity at the very base the section, could have been deposited at the end of chron C12n. Alternatively, it could correlate with Site LL21A from the Lima-Limo section, which Rochette *et al.* (1998) interpreted as excursional.

The GPTS of Ogg (2012) contains 14 chrons between 15 and 10 Ma, for a mean duration around 300 kyr per chron. The upper part of the Debre Sina section has no reverse polarities (except maybe one transitional site), indicating that it was probably emplaced in less than 300 kyr. With the existence of up to four chrons, the lower part of the Debre Sina section (Fig. 11) could span up to 1.2 Myr.

5 DISCUSSION

5.1 Emplacement of the Ethiopian traps

The correlation with the GTPS (Ogg 2012) indicates that the Oligocene Belessa section fully overlaps chrons C11n.2n and C11n.1r, partially chrons C11r and C11n.1n (Fig. 12). It implies that the section recorded a total duration between 0.59 and 1.40 Myr. As the polarity zones R1 (C11r) and N2 (C11n.1n) cover more than one half of the stratigraphy, it is realistic to consider the duration of the Belessa section to be about 1 Myr around 29–30 Ma. The respective emplacement rates for the successive polarity zones would be greater than 0.5 m kyr^{-1} for R1 (C11r), around 0.67 m kyr^{-1} for N1 (C11n.2n), around 1.1 m kyr^{-1} for R2 (C11n.1r), and greater

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Table 2. Palaeomagnetic results for the Debre Sina section: site (or group) name; longitude (slon), latitude (slat) and altitude (alt) of the site; stratigraphic height (h_{strat}); number n/N of interpreted/measured specimens by thermal (TH) or alternating field (AF) demagnetization; declination (D_{geo}) and inclination (I_{geo}) in geographic coordinates with the precision parameter (k) and the 95 per cent confidence radius (α_{95}); longitude (plon) and latitude (plat) of the virtual geomagnetic poles; median destructive field (MDF) and maximal unblocking temperature T_u. Flows marked with an asterisk (∗) were merged into directional groups highlighted in italics. Flows marked with a cross (†) record excursional directions.

Site/group	Slon [°E]	Slat [°N]	Alt [m]	h_{strat} [m]	n/N(TH)	n/N(AF)	D_{geo} [°]	I_{geo} [°]	k	α_{95} [°]	Plon [°N]	Plat [°E]	MDF [mT]	T_u [°C]
M04	39.7695	9.8457	2587	0	6/6	5/5	351.7	48.6	150	3.7	19.7	68.7	30	500
M07	39.7672	9.8460	2637	50	5/5	5/5	15.7	0.6	319	2.7	160.2	71.7	18	575
M06	39.7651	9.8461	2656	69	8/8	2/5	359.3	9.2	108	4.7	226.7	84.8	12	500
M46†	39.7592	9.8476	2709	122	2/3	5/9	308.7	−40.3	56	8.2	275.9	29.7	15	525
M48†	39.7588	9.8481	2724	137	0/3	8/8	65.6	−21.9	342	3.0	146.2	21.5	17	525
M51	39.7534	9.8429	2804	217	6/8	2/2	26.5	9.8	355	2.9	138.4	63.3	8	375
M08	39.7326	9.9370	2839	252	4/9	2/4	152.6	2.2	301	3.9	109.2	−60.6	9	400
M03	39.7314	9.9389	2861	274	6/6	3/5	17.0	11.2	122	4.7	142.6	72.6	13	600
M14∗	39.7305	9.9171	2877	290	6/6	2/2	350.5	4.3	131	4.9	271.0	77.7	26	575
M15∗	39.7305	9.9161	2881	294	3/3	8/10	351.0	5.2	325	2.5	271.0	78.4	23	575
M14–15				292	9/9	10/12	350.8	4.8	212	2.3	271.0	78.2		
M02†	39.7277	9.9419	2893	306	8/8	4/5	354.2	58.1	51	6.1	31.9	59.9	52	450
M16	39.7294	9.9152	2914	327	8/8	5/5	169.3	30.4	62	5.3	61.6	−61.4	30	575
M52	39.7544	9.8581	2951	364	3/3	7/7	18.8	−8.7	72	5.7	166.1	66.4	38	575
M53	39.7566	9.8588	2962	375	3/3	9/9	348.8	3.1	210	3.0	273.7	76.1	10	500
M54	39.7565	9.8588	2967	380	3/3	9/9	22.4	−15.6	124	3.9	167.6	61.4	58	575
M55	39.7581	9.8600	2977	390	3/3	7/8	349.7	2.9	66	6.0	271.1	76.7	25	525
M56	39.7560	9.8589	2991	404	3/3	6/8	0.6	24.2	326	2.9	50.2	87.1	23	575
M18†	39.7321	9.9057	2997	410	7/8	0/8	136.8	14.3	69	7.3	109.5	−43.7	4	425
M57	39.7540	9.8585	3000	413	3/3	8/9	7.0	11.6	424	2.2	159.1	82.0	12	500
M09	39.7330	9.9047	3009	422	3/3	8/8	356.0	33.0	824	1.6	14.8	81.0	25	575
M58	39.7522	9.8575	3016	429	3/3	8/8	357.4	9.2	190	3.3	246.5	84.2	20	525
M11	39.7389	9.8969	3025	438	3/3	10/10	4.8	0.1	176	3.1	193.4	79.0	26	550
M20	39.7384	9.8953	3064	477	2/3	8/8	11.4	21.3	95	5.0	122.8	78.7	12	500
M19	39.7388	9.8953	3068	481	4/7	2/8	355.9	13.2	220	4.5	271.6	84.8	13	450
M59	39.7480	9.8476	3072	485	3/3	8/8	9.7	−3.5	306	2.6	179.6	74.9	25	525
M22	39.7390	9.8939	3083	496	3/3	9/9	0.9	−0.8	94	4.5	214.5	79.6	7	450
M21	39.7403	9.8910	3098	511	3/3	8/9	344.1	11.3	222	3.1	296.1	73.7	13	525
M44	39.7423	9.8573	3114	527	2/3	8/8	349.3	27.3	390	2.4	335.0	78.6	9	450
M42	39.7413	9.8551	3135	548	3/3	8/8	0.1	11.3	316	2.6	218.6	85.8	25	525
M23	39.7348	9.8472	3159	572	3/3	8/8	7.0	4.6	227	3.0	176.6	79.8	5	475
M24	39.7356	9.8472	3170	583	3/3	10/10	3.5	14.1	341	2.2	167.5	85.7	8	425
M25	39.7364	9.8473	3181	594	4/8	1/2	340.5	10.6	145	6.4	298.2	70.2	6	400
M26	39.7379	9.8459	3202	615	3/3	9/10	9.2	5.9	91	4.6	166.2	78.6	7	375
M39	39.7447	9.8475	3202	615	2/3	4/8	16.5	−24.8	41	10.6	184.5	61.6	21	550
M28∗	39.7386	9.8457	3205	618	2/3	7/8	6.3	11.0	33	9.1	164.0	82.5	7	550
M35∗	39.7403	9.8432	3248	661	3/3	9/9	0.9	4.0	78	4.9	212.9	82.1	37	550
M28,M35				640	5/6	16/17	3.2	7.0	45	4.8	193.0	83.0		

than 1.36 m kyr^{-1} for N2 (C11n.1n), suggesting an acceleration in the volume of volcanism over time. The maximum duration of 1.40 Myr would correspond to a minimum emplacement rate of 0.7 m kyr^{-1}, reaching 1 m kyr^{-1} if one assumes a duration on the order of 1 Myr.

The dating of the Belessa section agrees with previous work on the Afro-Arabian traps (Hofmann *et al.* 1997; Rochette *et al.* 1998; Coulié *et al.* 2003; Riisager *et al.* 2005). The Belessa section, which samples the lower part of the traps, is half as thick as the *ca*. 2-km-thick Lima-Limo section (Rochette *et al.* 1998; Coulié *et al.* 2003), but better matches with the GPTS. For the Lima-Limo section, two interpretations are possible (Fig. 12): (1) Subchron C11n.1r was not recorded at Lima-Limo; (2) Subchron C11n.1r was recorded but not subchron C11n.1n. Both scenarios are geologically plausible as it has been recognized that the thickness of the Ethiopian traps can vary in space by more than a factor two (e.g. Mohr 1983).

5.2 Field behaviour during the Early Oligocene

The Belessa section can be compared with three other Early Oligocene sections in a 1000 km radius (Fig. 1): the Lima-Limo section located *ca*. 90 km farther north (Hofmann *et al.* 1997; Rochette *et al.* 1998), the Sanaa section in Yemen (Riisager *et al.* 2005), and the As Sarat section in Saudi Arabia (Kellogg & Reynolds 1983). The Belessa and Lima-Limo sections are continuous and can be directly compared because they are located on the same tectonic plate. The site-mean directions of the Belessa section do not pass the reversal test. The normal polarity population agrees to within 2–3° of the direction predicted by the 30.2 Ma synthetic pole for Africa (Besse & Courtillot 2002, 2003), whereas the population of reverse polarity is more than 10° away from the expected direction. As only 33 per cent of the stratigraphy consists of reverse

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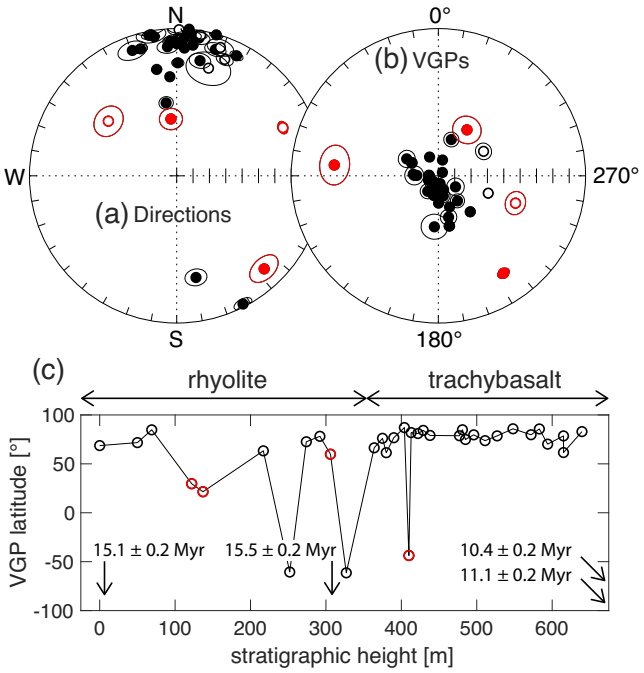


Figure 11. Site-mean directions for the Debre Sina section. Panel (a): dispersion of the directions. Panel (b): dispersion of the VGPs . Panel (c): magnetostratigraphy.

polarity, it could be that PSV was not sufficiently averaged for the reverse polarity sites. In contrast, the site-mean directions of the Lima-Limo section pass the reversal test (Rc classification of McFadden & McElhinny 1990), but the mean direction lies 6.5° away from that predicted by the 30.2 Ma synthetic pole for Africa. Rochette *et al.* (1998) explained the discrepancy by the presence of a 14 per cent axial quadrupole component. Riisager *et al.* (2005) disagreed with this interpretation because of its inconsistency with global Oligocene data. The Sanaa and As Sarat sections are composite sections, consisting of several spatially distant segments that were patched together, which does not permit a rigorous check for the serial correlation between consecutive flows. With an average of three samples per flow and blanket demagnetization, the As Sarat does not meet present-day palaeomagnetic standards. Therefore, we only retained the sites that have more than three samples and α_{95} values lower than 15°. This led to the retention of 37 out of 63 sites after applying the Vandamme (1994) cut-off. The Sanaa and As Sarat sections are located on the Arabian Plate, which additionally means that their directions may have been displaced from those of the Belessa and Lima-Limo sections located on the African Plate.

None of the four Oligocene sections is a perfect candidate to investigate PSV. It is furthermore desirable to have more than 100 time-independent determinations to calculate robust PSV estimates (e.g. Lhuillier & Gilder 2013). To circumvent this issue and negate effects from potential tectonic displacement between the African and Arabian plates, we recentred the directions and VGPs about their principal axes (e.g. Tauxe & Kent 2004; Lhuillier & Gilder 2013). We then combined the recentred populations of the four sections, which led to a set of 167 directions (Fig. 13a, Table 4). Fig. 13(a) shows that the distribution of directions is elongated in the meridional plane ($e = 2.7$), whereas the distribution of VGPs is nearly circular ($E = 1.1$), yielding $S = 14.2°|^{15.4°}_{13.2°}$. Removing the As Sarat data does not significantly alter these values (Table 4), so we retained them for further comparison.

We used a bootstrap approach to better assess the robustness of our estimates and determine the influence of the number N of directions/VGPs (e.g. Tauxe 2010). We considered subsets of the full data set with the possibility of repeating values for N ranging from 10 to 160. Fig. 14 shows the precision parameter k of the directions (panel a, circles), the VGP scatter S (panel a, squares), the elongation parameter e of the directions (panel b, circles) and the elongation parameter E of the VGPs (panel b, squares) as a function of N. Each of the four curves reveals that the considered parameter departs from its converged value at $N = 160$, by approximately 25 per cent at $N = 100$, and by approximately 10 per cent at $N = 130$. The analysis thus indicates that it is unlikely to determine precise estimates for $N \leq 100$, with a strong risk of a systematic bias for $N \leq 50$. Having more than 100 directions is practically difficult to obtain for a single study but can be attained by a combined data set. With $N \leq 100$ and a temporal window in excess of 2 Myr, we conclude that the Early Oligocene data set is likely to yield robust PSV estimates.

5.3 Field behaviour during the Middle Miocene

The Debre Sina section provides the only data set derived from volcanic rocks in a 1000 km radius of the Belessa section to investigate PSV during the Middle Miocene. The Turkana basalts of northwestern Kenya (Raja *et al.* 1966; Reilly *et al.* 1976) were initally claimed to be Middle Miocene, but recent studies have since dated most of them as Eocene or Oligocene (e.g. Brown & McDougall 2011). The only well-dated magnetic records that can be used for comparison with the Debre Sina section are sediments of the Ngorora Formation from the East African rift (10–13 Ma, Tauxe *et al.* 1985; Deino *et al.* 1990). As they were emplaced at low latitude, we assume that inclination shallowing was negligible. Deino *et al.* (1990) reported the results of four partially overlapping continuous sections that we combined to produce a data set of 95 directions after applying the Vandamme (1994) cut-off.

We again recentred the directions and VGPs about their principal axes. We then combined the recentred populations of the Debre Sina section and the Ngorora Formation, which led to a set of 125 directions (Fig. 13b, Table 4). The VGP scatter $S = 15.0°|^{16.5°}_{13.8°}$ is indistinguishable at the 95 per cent significance level from $S = 14.2°|^{15.4°}_{13.2°}$ for the Early Oligocene (Fig. 15). The elongation parameter $e = 1.8 \pm 0.6(2\sigma)$ is 30 per cent lower than $e = 2.7 \pm 0.8(2\sigma)$ for the Early Oligocene. This last observation is subject to caution due to the composite nature of the Middle Miocene data set. Bootstrap experiments however indicate that the results from the Middle Miocene data set are converged (Figs S3a and b).

5.4 Field behaviour during the Plio-Pleistocene

We selected seven studies that reported mean directions from 23 to 66 independent lava flows to investigate field behaviour during the Plio-Pleistocene. In the Afar region, the data sets from the Gamarri (Kidane *et al.* 1999) and Dobi (Ahn *et al.* 2016) cliffs are continuous sections, whereas the collection from the Alayta-Dabbahu magmatic segment (Muluneh *et al.* 2013) consists of scattered lava flows. In the Ethiopian part of the East-African rift region, only one study near the Kereyou lodge (Kidane *et al.* 2010) is a continuous section, whereas two other studies on the Fentale segment (Kidane *et al.* 2009) and on the Dofan segment (Nugsse *et al.* 2018) consist of

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Table 3. Unspiked K-Ar ages for the Belessa and Debre Sina sections. The ages in bold were retained for interpretation.

Sample	Weight molten	K∗	^{40}Ar∗	^{40}Ar∗	^{40}Ar∗ weighted mean ±1σ	Age ±2σ
Experience	[g]	[weight %]	[%]	[10^{-11}mol/g]	[10^{-11}mol/g]	[Ma]
Belessa section						
B02 (12.4446°N, 37.7825°E, 1860 m)						
252	0.30122	0.390±0.004	58.754	1.875	1.865±0.005	27.36±0.40
262	0.53475	0.390±0.004	59.215	1.873		
268	0.34750	0.390±0.004	59.901	1.847		
B06 (12.4499°N, 37.7786°E, 1885 m)						
280	0.29750	0.423±0.004	61.350	2.035	2.031±0.007	27.48±0.41
299	0.28742	0.423±0.004	60.517	2.027		
B68 (12.4416°N, 37.6611°E, 2610 m)						
287	0.25260	0.656±0.007	83.911	3.418	3.392±0.010	**29.57±0.44**
294	0.20645	0.656±0.007	81.794	3.355		
300	0.22363	0.656±0.007	82.675	3.403		
Debre Sina section						
M05 (9,8458°N, 39,7697°E, 2591 m)						
391	0.29963	3.918±0.039	84.893	10.333		**15.08±0.23**
					10.292±0.036	
393	0.30333	3.918±0.039	83.925	10.252		
M02 (9.9419°N, 39,7277°E, 2893 m)						
368	0.30869	3.595±0.036	96.248	9.787	9.672±0.035	**15.45±0.23**
384	0.29826	3.595± 0.036	96.136	9.563		
M33 (9.8436°N, 39,7421°E, 3300 m)						
366	0.31199	1.718±0.018	85.864	3.342	3.320±0.012	**11.11± 0.17**
372	0.30783	1.718±0.018	81.794	3.299		
M34 (9.8433°N, 39.7422°E, 3308 m)						
375	0.29755	1.718±0.018	84.914	3.121	3.110±0.011	**10.40±0.16**
383	0.30408	1.718±0.018	85.956	3.099		

scattered lava flows. Further south, Opdyke *et al.* (2010) investigated two collections of scattered lava flows from Mount Kenya near the Equator and from the Loiyangalani region in the Kenyan rift at 3°N.

We combined the recentred populations of the seven studies, which led to a set of 249 directions after the application of the Vandamme (1994) cut-off. Fig. 13(c) shows that the recentred directions are elongated in the meridian plane. To within 95 per cent uncertainty estimated by bootstraping, the elongation parameter $e = 2.4 \pm 0.8(2\sigma)$ is not significantly different from $e = 2.6$ predicted by the TK03.GAD statistical model for the past 5 Myr (Tauxe & Kent 2004). In contrast, the VGP scatter $S = 9.6°|^{10.2°}_{9.0°}$ is approximately 15 per cent lower than the value $S = 11.0°|^{11.4°}_{11.2°}$ predicted by TK03.GAD. Bootstrap experiments confirm that the results from the Plio-Pleistocene data set are converged and corroborate the previous statement that a collection of more than 100 directions is desirable to derive robust PSV estimates (Figs S3c and d).

5.5 Comparison between the three epochs

For the three investigated epochs (Early Oligocene, Middle Miocene and Plio-Pleistocene), the elongation in directions or VGPs is consistent within uncertainty with that predicted by the TK03.GAD statistical model for the past 5 Ma (Table 4, Fig. 13). Considering the elongation parameter as a proxy of the geometry of the field (e.g. Tanaka 1999), we conclude that the field was probably as dipolar during the Early Oligocene and the Middle Miocene as it was during the past 5 Myr. In contrast, the VGP scatter S during the Early Oligocene and the Middle Miocene was 45–55 per cent higher than during the Plio-Pleistocene (Table 4, Figs 13 and 15). As the field exhibits a comparable geometry for the three investigated epochs indicated by the consistent elongation in directions, the 45–55 per cent decrease of the VGP scatter can also be seen as a 55–60 per cent increase of the precision parameter k in directions (Table 4, Figs 13 and S4).

The commonly used Vandamme (1994) cut-off procedure enables one to rigorously compare the directions of stable polarity. However, the discrepancy in VGP scatter between the Early Oligocene–Middle Miocene and the Plio-Pleistocene remains, albeit reduced, even when the cut-off procedure is not applied (Table S2, Fig. S4). Another critical point when comparing PSV between various epochs is to ensure that each data set sampled a sufficient amount of time to produce converged statistics. This is likely for the Early Oligocene data set that covers the *ca*. 2 Myr history of the Afro-Arabian traps (Riisager *et al.* 2005). It is also likely for the Plio-Pleistocene data set, noting that the study from Mount Kenya included approximately 5 Myr (Opdyke *et al.* 2010) and yielded comparable S values to those from Ethiopia. Since both Early Oligocene and Plio-Pleistocene data sets span several million years, it is thus likely that the difference in S observed between these two epochs is a robust feature and does not stem from a sampling bias. Despite a duration of several millions of years, the Middle Miocene data set is less well constrained because it is only relies on two studies and one study was based on sediments (Deino *et al.* 1990) making the comparison with volcanic rocks ambiguous.

That VGP scatter during the Early Oligocene–Middle Miocene was approximately 50 per cent higher than during the Plio-Pleistocene thus appears to be a robust feature in the Ethiopian area. The geomagnetic polarity timescale (Ogg 2012) shows that reversal frequency was comparable between the Plio-Pleistocene

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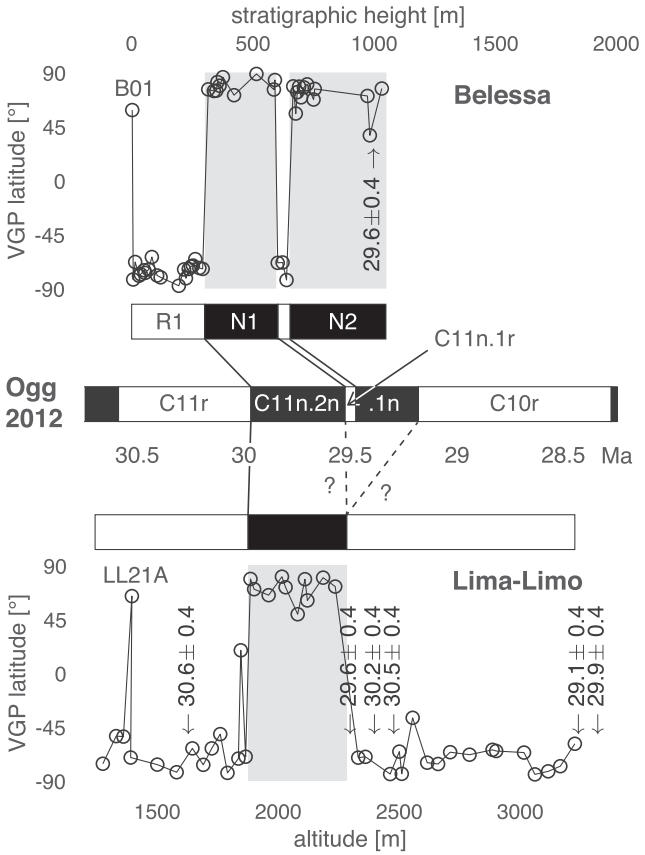


Figure 12. Comparison of the Belessa and Lima-Limo sections with the geomagnetic polarity timescale by Ogg (2012). Magnetostratigraphy from Rochette *et al.* (1998) for the Lima-Limo section with K-Ar ages from Coulié *et al.* (2003). The dashed lines correspond to the two possible interpretations for the end of the normal polarity zone (see Section 5.1 for details).

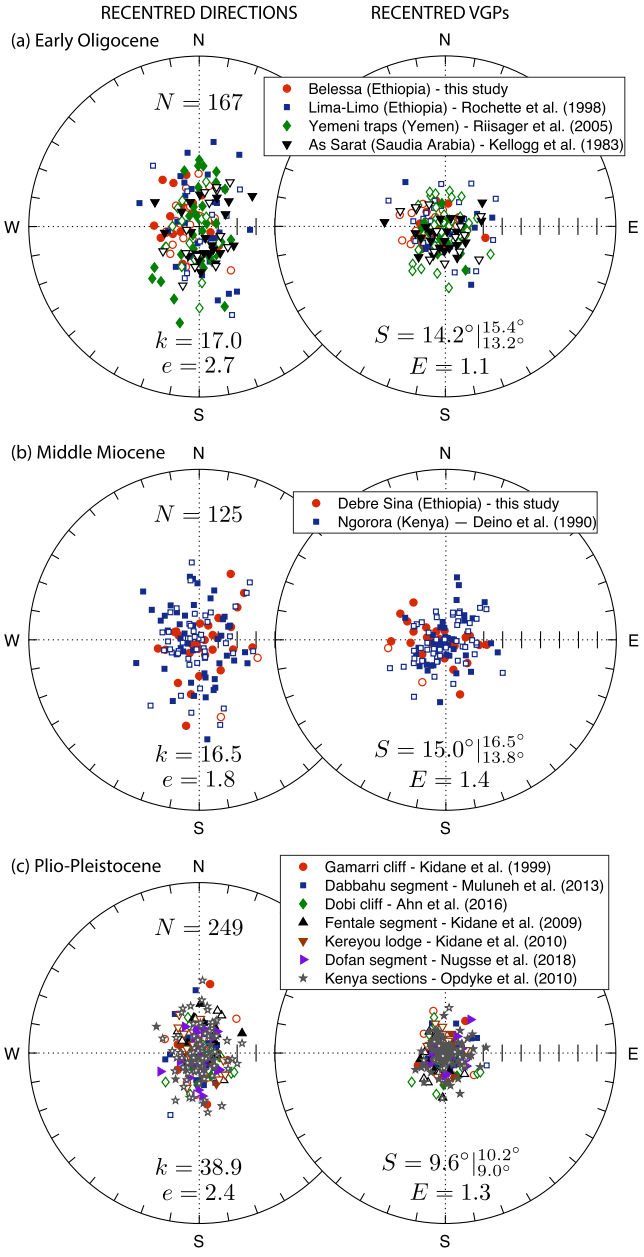


Figure 13. Comparison of the field behaviour during the Early Oligocene (first row), the Middle Miocene (second row) and the Plio-Pleistocene (third row). The left (resp. right) stereonets show the distribution of the recentred directions (resp. of the recentred VGPs). The quantities k and e stand for the precision and elongation parameters of the recentred directions, the quantities S and E for the angular standard deviation and elongation parameter of the recentred VGPs.

(4.9 Myr^{-1}) and the Middle Miocene (5.7 Myr^{-1}), whereas reversal frequency was at least twice lower during the Early Oligocene (1.6 Myr^{-1}). It is thus seemingly contradictory that VGP scatter would be twice higher during the Early Oligocene than during the Plio-Pleistocene (Fig. 15). Palaeomagnetic data comparing the geomagnetic field behaviour during the Cretaceous Normal Superchron and the past 5 Myr (e.g. Biggin *et al.* 2008) together with numerical dynamo simulations (e.g. Lhuillier & Gilder 2013) suggest a positive correlation between reversal frequency and PSV. In contrast, the data from the Ethiopian area during the past 30 Myr suggest that the dependency between reversal frequency and PSV may be more complicated. It remains an open question whether the apparent decoupling between S and f is ascribable to a local anomaly, is only sporadic in time, or represents a general feature of the Earth's magnetic field.

6 CONCLUSIONS

Due to an abundance of volcanic rocks, Ethiopia is a choice place to investigate palaeosecular variation at low latitude during the past 30 Ma. We studied two new sequences of continuous lava flows that, combined with previously published results, leads to the following conclusions:

(i) The distribution of recentred directions are elongated in the meridional plane, which points to a persistent axial dipolar field over the past 30 Myr.

(ii) The dispersion S of the VGPs, which quantifies PSV, is approximately 50 per cent higher during the Early Oligocene–Middle Miocene than during the Plio-Pleistocene.

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Table 4. Mean directions (declination ⟨*D*⟩, inclination ⟨*I*⟩) and VGPs (longitude ⟨plon⟩, latitude ⟨plat⟩) for the Belessa and Debre Sina sections compared to results from other studies: *N*, number of flows/determinations; *k* (or *K*), precision parameter; a_{95} (or A_{95}), 95 per cent confidence radius; *e* (or *E*), elongation parameter; *S*, angular dispersion of VGPs with it 95 per cent confidence interval. Rev. test indicates the classification of the reversal test according to McFadden & McElhinny (1990). BC02 refers to the synthetic pole for Africa (Besse & Courtillot 2002, 2003), TK03.GAD and MM97 to the field models of Tauxe & Kent (2004) and McElhinny & McFadden (1997).

Section	Directions			VGPs											Rev. test
	N	⟨*D*⟩	⟨*I*⟩	*k*	a_{95}	*e*	⟨plon⟩	⟨plat⟩	*K*	A_{95}	*S*			*E*	
Early Oligocene sections															
Belessa	42	356.4	7.1	26.7	4.3	1.6	240.6	80.5	44.9	3.3	10.5	12.1	14.3	2.9	R–
Belessa normal	20	2.1	5.6	30.0	6.1	2.7	205.8	80.3	61.3	4.2	8.4	10.3	13.3	1.5	
Belessa reverse	22	351.2	8.4	29.5	5.8	1.9	265.9	78.1	50.6	4.4	9.4	11.4	14.5	3.0	
BC02 (30.2 Ma)	13	1.9	3.1				207.7	79.0	63.3	5.4					
Lima-Limo	40	2.5	−1.6	11.7	6.9	3.3	207.2	75.8	24.0	4.7	14.4	16.6	19.7	1.5	Rc
Lima-Limo normal	11	0.1	−1.6	10.5	14.8	3.2	217.6	76.2	21.0	10.2	13.6	17.8	25.7	1.6	
Lima-Limo reverse	29	3.4	−1.6	11.7	8.2	3.4	203.4	75.5	24.7	5.5	13.8	16.4	20.1	1.4	
BC02 (30.2 Ma)	13	1.9	4.8				207.7	79.0	63.3	5.4					
Sanaa	48	353.5	1.4	14.4	5.6	4.9	248.8	74.0	33.1	3.6	12.4	14.2	16.5	1.8	Rc
Sanaa normal	15	356.2	7.1	21.8	8.4	4.5	243.0	77.9	51.1	5.4	9.0	11.4	15.4	1.4	
Sanaa reverse	33	352.2	−1.2	12.9	7.2	5.3	250.7	72.2	29.3	4.7	12.8	15.0	18.2	2.0	
As Sarat	37	354.4	15.6	22.2	5.1	2.1	247.0	75.9	33.5	4.1	12.1	14.1	16.8	1.5	Rc
As Sarat normal	23	356.4	17.9	26.2	6.0	1.4	241.7	78.6	42.2	4.7	10.4	12.5	15.8	2.6	
As Sarat reverse	14	351.2	11.7	18.5	9.5	5.2	252.3	71.1	27.7	7.7	12.2	15.5	21.2	2.5	
combined (with AS)	167			17.0	2.7	2.7			32.9	1.9	13.2	14.2	15.4	1.1	Rb
combined (without AS)	130			15.8	3.2	3.0			32.5	2.2	13.1	14.3	15.6	1.0	Rb
Middle Miocene sections															
Debre Sina	30	1.5	7.1	16.6	6.6	1.8	206.9	83.8	27.7	5.1	13.1	15.5	18.9	2.4	Ro
Debre Sina normal	28	2.9	8.7	19.8	6.3	2.6	191.1	84.1	32.7	4.8	12.0	14.2	17.5	3.4	
Debre Sina (<2950 m)	8	0.9	6.6	8.7	20.0	1.6	214.4	83.9	13.8	15.4	16.1	22.0	34.6	2.0	
Debre Sina (>2950 m)	22	1.7	7.3	22.4	6.7	3.5	204.4	83.7	38.8	5.0	10.7	13.0	16.6	4.4	
BC02 (11.92 Ma)	21	3.8	12.9				170.7	85.0	107.4	3.1					
Ngorora	95	356.0	−4.1	16.3	3.7	1.9	266.6	84.9	29.7	2.7	13.6	15.0	16.7	1.8	Rb
Ngorora normal	47	357.3	−3.1	17.6	5.1	2.1	262.5	86.2	32.0	3.7	12.6	14.4	16.8	1.9	
Ngorora reverse	48	354.8	−5.1	15.1	5.5	2.1	268.9	83.6	27.5	4.0	13.6	15.5	18.1	2.0	
combined	125			16.5	3.2	1.8			29.4	2.4	13.8	15.0	16.5	1.4	Rb
Plio-Pleistocene sections															
Gamarri	28	8.3	8.9	37.7	4.5	2.5	171.4	79.2	74.8	3.2	7.9	9.4	11.6	1.7	Rc
Dabbahu	33	356.0	11.8	39.3	4.0	2.7	252.0	82.3	81.9	2.8	7.7	9.0	10.9	1.5	Ro
Dobi	27	8.0	12.7	21.3	6.2	3.1	162.1	81.1	37.0	4.6	11.2	13.4	16.6	1.5	Rc
Fentale	26	353.3	19.5	39.0	4.6	2.4	321.8	83.1	68.3	3.5	8.2	9.8	12.2	2.1	R-
Kereyou	24	355.3	12.9	25.0	6.0	4.1	283.5	84.9	51.5	4.2	9.4	11.3	14.2	2.6	Rc
Dofan	23	354.1	13.2	44.7	4.6	3.1	287.4	83.6	92.5	3.2	7.0	8.4	10.7	1.3	N/A
Mount Kenya	66	1.6	−0.6	32.3	3.1	2.3	141.3	88.4	65.6	2.2	9.0	10.0	11.4	1.6	Rb
Loiyangalani	30	0.8	−0.8	38.5	4.3	2.3	203.4	86.7	76.4	3.0	7.9	9.3	11.4	1.8	Rb
BC02 (2.1 Ma)	25						182.2	86.6	96.2	3					
combined	249			38.9	1.4	2.4			71.8	1.1	9.0	9.6	10.2	1.3	Ra
PSV models at 10°N															
TK03.GAD				15.1		2.6					11.0	11.2	11.4		
MM97											11.2	12.3	13.6		

(iii) As the reversal frequency *f* was more than twice lower during the Early Oligocene than during the Plio-Pleistocene, the two proxies (*f* and *S*) might be, locally or at times, uncorrelated.

From a methodological point of view, we recall that

(i) The quasi-Fisherian distribution of VGPs from low latitude sites is more appropriate to estimate PSV than the distribution of directions.

(ii) The elongation parameter *e* of the directions may be an efficient tool to characterize field morphology.

(iii) A robust estimate of *e* requires more than 100 site-mean directions, which is difficult to achieve in a single study but can be attained by combining data sets from multiple studies.

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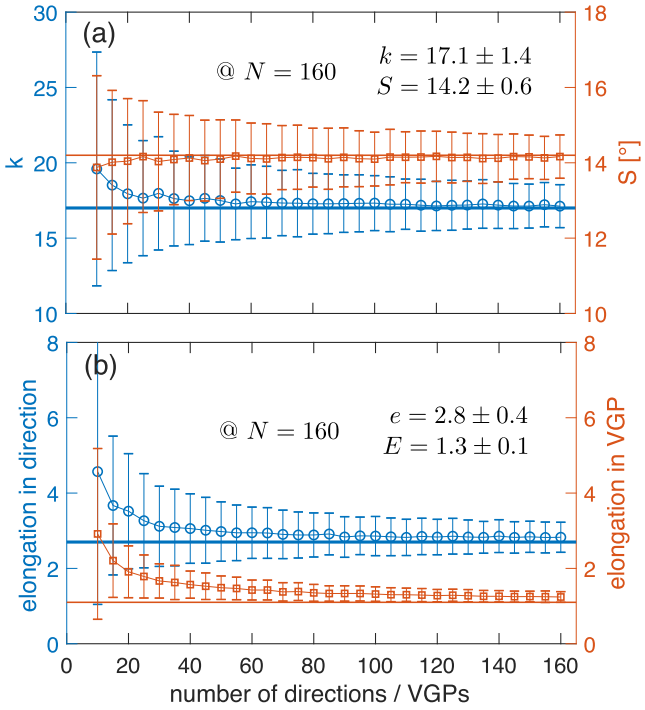


Figure 14. Bootstrap experiments on the combined Early Oligocene data set to determine the precision parameter k of the directions (panel a, circles), the angular dispersion S of the VGPs (panel a, squares), the elongation parameter e of the directions (panel b, circles), and the elongation parameter E of the VGPs (panel b, squares). For each increment in the number N of directions/VGPs, the average value computed from 1000 random draws is shown with its 1σ-dispersion. The thick (resp. thin) lines represent the values of k and e (resp. S and E) for the whole data set.

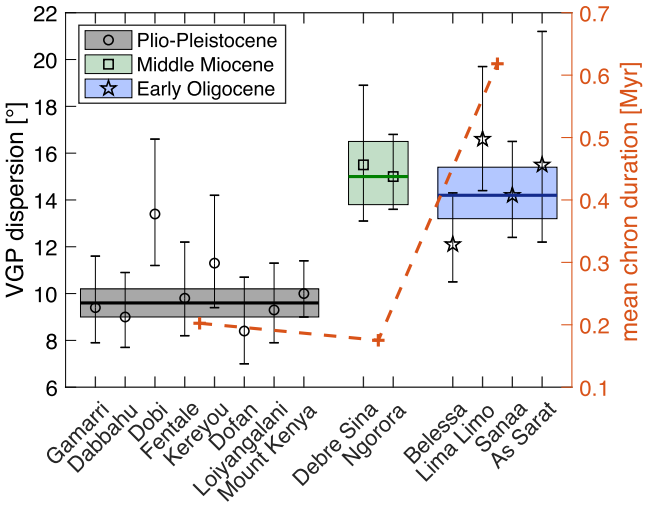


Figure 15. Comparison of the VGP dispersion (open symbols) and the mean chron duration as a proxy for reversal frequency (dashed line) during the Plio-Pleistocene, the Middle Miocene and the Early Oligocene. Detailed data are reported in Table 4.

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SUPPORTING INFORMATION

Supplementary data are available at *GJI* online.

Table S1. Chemical analyses of the major elements.
Table S2. Directional analysis without Vandamme (1994) cut-off
Figure S1. Thermomagnetic experiments for the Belessa section
Figure S2. Thermomagnetic experiments for the Debre Sina section
Figure S3. Bootstrap experiments on the combined Middle Miocene and Plio-Pleistocene data sets
Figure S4. VGP scatter *S* as a function of the precision parameter *k* for directions

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